COMPUTER N/W SHORT NOTES

- IP address is the address of the conn^n not the host. If a device has two connection to the Internet, via two n/w, it has two IPV4 addresses.
- IPV4 32 bit address, divided into 5 classes ← Address Space $= 2^{32}$

Class	Fixed Part	Range	IP add	Host	N/W
A	0	1-126	2^{24}	$2^{24}-2$	2^{7}
B	10	128-191	2^{16}	$2^{16}-2$	2^{14}
C	110	192-223	2^{8}	$2^{8}-2$	2^{21}
D	1110	224-239	–		
E	1111	240-255	–		

-2 bcz 1st IP is used to represent n/w and last is broadcast IP

- 0.0.0.0 – DHCP Client
- 127.x.y.z – loop back address.

Quick Binary maths

$01111111 = 2^7-1 = 128-1 = 127$

$00111111 = 2^6-1 = 64-1 = 63$ etc...

10000000 → 128	11111000 → 248
11000000 → 192	11111100 → 252
11100000 → 224	11111110 → 254
11110000 → 240	11111111 → 255

- In Routing, only destination IP matters.

Limited Broadcast

255.255.255.255

(X) Router → Blocked

Local n/w only (192.168.1.0/24)

PC1 .10 | PC2 .20 | PC3 .30

✗ Never crosses routers

✓ same subnet only

↓ shouting in a room, where door is closed

Direct Broadcast

192.168.1.255

(X) Router → Forwarded

Specific N/W (192.168.1.0/24)

PC1 .10 | PC2 .20 | PC3 .30

✓ can cross routers

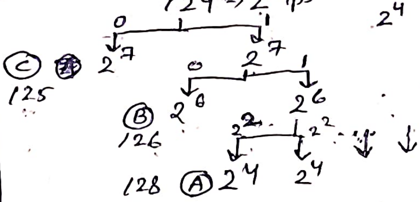
✓ target specific n/w

↓ addressing a letter to everyone at a specific address

- Subnets are borrowed from host id, and ~~ca~~ not always be contiguous. (Not necessary)
  12.0.3.75/.
- Subnet mask → part of N/W id → put 1 in all places.
- N/W id → host part will be 0.
- Whenever we have to divide subnets of different no. of hosts use huffman encoding method. Also, the slash notation will be $\boxed{32 - \log_2(2^{\text{bits req to suffice host}}) \Rightarrow 32 - \text{bits req}...}$

Eg:- /24 into 3 n/w of 10, 60, 120 ips.
(A) (B) (C)

$/24 \Rightarrow 2^8$ ips; 2^4, 2^6, 2^7

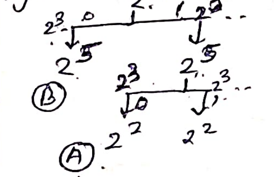


Huffman encoding gives bits after actual slash notation

Range (A) = $\underbrace{x.y.z.1100}_{\text{Fixed}}0000$ to $\underbrace{x.y.z.1100}_{\text{Fixed}}1111$

= x.y.z.192/28 to x.y.z.207/28

Eg 2: 2^8; $\frac{2^8}{2^5} = 2^3$; $\frac{2^5}{2^2} = 2^3$



Don't forget to write the pattern from the parent

∴ A = x.y.z.001000 00 to x.y.z.001000 11

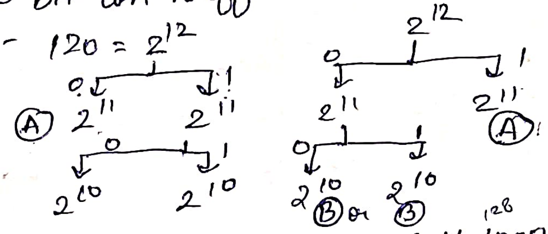
B = x.y.z.000 00000.00000000 to x.y.z.000 11111.11111111

- If subnet mask is 255.255.31.0 then first two octets should be as it is (no change)
- Historically (older cisco rules):-
  (i) subnet with all 0's all 1's ⇒ reserved
  (ii) usable subnets $= 2^{\text{bits of subnet}} - 2$

Modern practice :-
(i) All subnets are usable (including 0 & all 1's)

(i) It is not necessary to always allocate 0 bit in huffman encoding.

Eg:- $/20 = 2^{12}$



$A = \underbrace{x.y.1000}_{\text{Fixed}}\, \underbrace{0}_{\text{from huffman}}0...$ = x.y.128.0

$A = \underbrace{x.y.1000}_{\text{Fixed}}\,\underbrace{1}_{\text{from huffman}}000...$ = x.y.136.0

- Router only care for destination IP
- We can only combine n/w if they are routed to same interface or router
- If more than ~~one~~ one entry matches in the routing table, then choose the one which has the largest (in no.) slash notation
- Have a separate entry which we can't combine

- If we want to combine some n/w just check till what point things are matching
- If after aggregation we cover extra IPs then mention that entry separately and our largest prefix match will take care i.e entry only when they are allocated to someone.
- <u>Misconceptions</u>

(i) addresses should be contiguous
(ii) No of subnets to be combined should be in 2^k

- supernet bits are borrowed from n/w id
- $\boxed{\text{No. of subnets} = \text{No of interfaces}}$
- when <u>two routers connect</u>, they use a <u>shared subnet</u> for their connection

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Points to remember:- (Supernetting)

(i) The first address of the block should be divisible by no of addresses of the block.

Eg:- 20.30.40.160
Block contain 32 IP addresses, which is first address of block.

Ans $32\overline{)160}(5$ ✓ so 20.30.40.160 can be first IP of block
$\underline{160}$
$\times$